Client Requirement Specifications

Supermarket Management System

Client: Nanoomal Jagmeshwar Dayal

Group Number: 23



Bachelor of Technology
Computer Science and Engineering

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Requirement Specification

Nanoomal Jagmeshwar Dayal operates a wholesale dealership based in Muzaffarnagar (U.P.) specializing in Kriyana, Dryfruits, Desi Jadi Booti, and Tota Rang Gulal. The management intends to replace its existing manual inventory and ledger processes with a centralized wholesale database management system. The proposed system should enable the business to efficiently manage stock levels, calculate accurate GST slabs, monitor customer demand trends, and allow administrators to manage suppliers, employees, and inventory seamlessly.

Customer Information

The system needs to keep track of every regular bulk buyer who registers. Each customer gets a unique Customer ID. Name is stored as a composite attribute made up of First Name and Last Name. Address is also handled as a composite attribute broken down into Shop/Building Number, Street, Area, City, State, and PIN Code. The system also records their Phone Number and accumulated Loyalty Points.

Employee Information

The business employs several staff members, each uniquely identified by an Employee ID. Along with the ID, the system stores their Name, Phone Number, Shift Schedule, and the specific item category they are assigned to manage.

Category Information

In the wholesale facility, inventory is grouped into distinct categories (such as Kriyana, Dryfruits, and Desi Jadi Booti). Each category has a unique Category ID, a Name, and a designated physical storage location or aisle number.

Item Information

Every product in the system has a unique Item ID. Along with the ID, the system stores the Item Name, Wholesale Price, Current Live Stock Level, HSN Code, and an Expiry Date. The actual GST tax percentage is not stored directly; rather, it is a derived attribute calculated on the fly based on the item's mapped HSN Code during billing.

Supplier Information

The dealer procures items from various distributors. Each supplier is identified by a unique Supplier ID, Name, Phone Number, and GSTIN (such as 09AAAFN7008D1ZL). This enables the system to track inward purchases and trigger restocking alerts. Supplier addresses are also stored as composite attributes.

Bill & Transaction Information

Every checkout transaction is recorded using a unique Bill ID. This record holds the associated Customer ID, Employee ID (Cashier), Transaction Date and Time, Payment Method (Cash, Card, UPI), and the Final Total Amount. Since one bill can cover multiple different items, the exact items and quantities purchased are treated as a multivalued relationship managed via a junction entity (Bill Details).

Functional Requirements

Authorized staff members and administrators can log into the system. Once logged in, they are able to:

- Browse all available inventory and check live stock levels across categories.
- Search for items by name, category, or HSN code.
- Process customer bills with automated GST calculation based on item HSN codes.
- Log unfulfilled customer stock requests for items currently out of stock.

Whenever a bill gets confirmed, the current live stock of the purchased items is automatically deducted. On the admin side, the administrator manages the system by adding or removing items, updating wholesale prices, managing supplier records, and analyzing customer purchase histories to generate targeted marketing lists.

Non-Functional Requirements

The user interface should be simple and intuitive enough that staff members can use it during fast-paced billing without difficulty. Data integrity and consistency need to be maintained at all times, ensuring live stock levels do not fall into negative values during concurrent transactions. Search and retrieval of inventory data should be highly responsive, automated GST calculations must be precise, and the overall system should be scalable enough to handle a growing catalog of wholesale products.